# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

## Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 2 – Case Study

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|---|---|---|---|
| **Course Code:** | DSA03 | **Course Title:** | Natural Language Processing |
| **CO Assessed:** | CO3 – Precision | **Assessment:** | Assessment Tool 2 – Case Study |
| **Weightage:** | 30% of CIA | | |
| **CO5 Statement:** | Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3) | | |
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| **Section / Batch:** | BTECH AI & DS | **Date:** | 11-08-2026 |

### Code of Conduct

*I* Renee Vincent ASR-192424161 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Renee Vincent

### Instructions

- Read all three case studies carefully before attempting the questions.
- Provide appropriate justifications and interpretations for every computed result.
- Use NLP concepts, algorithms, and terminology wherever applicable.
- Diagrams, flowcharts, tables, or mathematical formulations may be used to support your answers.
- Duration: 30 minutes.

| Section / Topic | Focus | Case Study |
|---|---|---|
| N-Gram Language Models and Smoothing Techniques | Bigram and Trigram probability computation, Maximum Likelihood Estimation (MLE), Backoff models, Deleted Interpolation, Entropy calculation, uncertainty analysis, real-time next-word prediction | Case Study 1 – Smart Mobile Keyboard Prediction System |
| Part-of-Speech (POS) Tagging and Word Classes | Penn Treebank tagset, contextual ambiguity resolution, rule-based tagging, stochastic tagging (HMM), emission and transition probabilities, word class identification, chatbot language understanding | Case Study 2 – AI-Powered Customer Support Chatbot |
| Transformation-Based Tagging (Brill Tagger) | POS tag correction, transformation rules, contextual tagging, linguistic validation, tag sequence refinement, ambiguity handling | Case Study 3 – News Analytics and POS Tag Correction System |
| Corpus Statistics and Probabilistic Analysis | Word frequency counting, corpus-based probability estimation, entropy-based uncertainty measurement, tagging confidence evaluation | Case Study 3 – News Analytics and POS Tag Correction System |
| Integrated Syntactic Analysis and NLP Applications | Application of language models, POS tagging, probabilistic methods, corpus analysis, ambiguity resolution, and decision-making in real-world NLP systems | Case Studies 1, 2, and 3 |

## Annexure A – Case Study

### CASE STUDY 1: Smart Mobile Keyboard Prediction System

A smartphone company is developing an AI-powered keyboard application that predicts the next word while users type emails, messages, and social media posts. The language model is trained on the corpus:

*"Data science is powerful, data science drives innovation, data science is evolving."*

The development team employs Bigram and Trigram Language Models, Backoff Models, Deleted Interpolation, and Entropy Analysis to improve prediction accuracy and handle unseen word sequences.

The following statistics are obtained from the corpus:

- C(data) = 3
- C(data science) = 3
- C(science is) = 2
- C(science drives) = 1
- C(science drives innovation) = 1

The interpolation weights are:

- $\lambda_1 = 0.5$ (Trigram)
- $\lambda_2 = 0.3$ (Bigram)
- $\lambda_3 = 0.2$ (Unigram)

The prediction probabilities after the word "science" are:

- P(is) = 0.66
- P(drives) = 0.33

### Questions

1. Compute the Maximum Likelihood Estimate (MLE) of the Bigram Probability P(science | data). Interpret how this probability influences next-word prediction in a mobile keyboard application.
2. A user types the unseen sequence "data science improves". Apply the Backoff Model and estimate the probability using lower-order n-grams. Explain why backoff is essential for real-time predictive text systems.
3. Using Deleted Interpolation, calculate the probability of the sequence "data science is". Discuss how interpolation balances reliability and coverage in sparse datasets.
4. Calculate the Entropy of the distribution P(is)=0.66 and P(drives)=0.33. Interpret

the result and evaluate how entropy helps measure uncertainty and prediction confidence in intelligent keyboard systems.

## 1. Maximum Likelihood Estimate (MLE)

### Given

C(data) = 3
C(data science) = 3

### Formula

P(science | data) = C(data science) / C(data)

### Calculation

P(science | data) = 3 / 3
P(science | data) = 1.0
P(science | data) = 100%

### Answer

The MLE of the bigram probability P(science | data) is **1.0 or 100%**.

### Interpretation

The word "science" occurs after "data" three times, and the word "data" also occurs three times in the corpus. Therefore, every observed occurrence of "data" is followed by "science". The language model therefore assigns a probability of 1.0 to "science" after "data". In a mobile keyboard, this means that when the user types "data", the system can confidently suggest "science" as the next word. However, this probability is completely dependent on the training corpus and may not work well for unseen or different contexts.

## 2. Backoff Model

### Given

User input:
"data science improves"
The required trigram is:
P(improves | data, science)
The sequence "data science improves" does not occur in the training corpus.
Therefore:
P(improves | data, science) = 0
Since the trigram is unavailable, the model backs off to the bigram:
P(improves | science)
The bigram "science improves" is also not present.
Therefore:

P(improves | science) = 0
The model finally backs off to the unigram:
P(improves)
The word "improves" does not occur anywhere in the given corpus.
Therefore:
P(improves) = 0

**Final Answer**

P(improves | data, science) = 0
**Interpretation**

The backoff model works by using the highest-order N-gram available. It first checks the trigram, then the bigram, and finally the unigram. This approach is important in real-time predictive text because users frequently type new or uncommon word combinations that may not have appeared in the training corpus. Backoff prevents the system from depending only on exact longer sequences. In this particular example, the final probability is still zero because the word "improves" is completely absent from the corpus. In a practical keyboard, smoothing would normally be combined with backoff to give unseen words a small non-zero probability.

**3. Deleted Interpolation**

**Given**

Interpolation weights:
$\lambda 1 = 0.5$ (Trigram)
$\lambda 2 = 0.3$ (Bigram)
$\lambda 3 = 0.2$ (Unigram)
The required sequence is:
"data science is"

**Step 1: Trigram Probability**

The sequence "data science is" occurs 2 times.
The bigram "data science" occurs 3 times.
Therefore:
P(is | data, science) = C(data science is) / C(data science)
P(is | data, science) = 2 / 3
P(is | data, science) = 0.667

**Step 2: Bigram Probability**

The sequence "science is" occurs 2 times.
The word "science" occurs 3 times.
Therefore:

P(is | science) = C(science is) / C(science)
P(is | science) = 2 / 3
P(is | science) = 0.667

**Step 3: Unigram Probability**

The total number of words in the corpus is 18.
The word "is" occurs 2 times.
Therefore:
P(is) = 2 / 18
P(is) = 0.111

**Step 4: Interpolation Calculation**

Formula:

P(is | data, science) =
$\lambda 1$ P(is | data, science)

- $\lambda 2$ P(is | science)
- $\lambda 3$ P(is)

Substituting the values:
= (0.5 × 0.667) + (0.3 × 0.667) + (0.2 × 0.111)
= 0.3335 + 0.2001 + 0.0222
= 0.5558

**Final Answer**

P(is | data, science) ≈ **0.556**
or
**55.6%**

**Interpretation**

Deleted interpolation combines information from three levels: trigram, bigram, and unigram. The trigram receives the highest weight of 0.5 because it provides the most specific context. The bigram receives 0.3 and provides useful information when trigram evidence is weak. The unigram receives 0.2 and provides general word-frequency information. This combination improves prediction reliability and reduces the problem of sparse data. Therefore, interpolation provides a balance between contextual accuracy and coverage.

**4. Entropy Analysis**

**Given**

P(is) = 0.66

P(drives) = 0.33

**Formula**

$H = -\Sigma P(x) \log_2 P(x)$
For two possible next words:
$H = -[P(is)\log_2 P(is) + P(drives)\log_2 P(drives)]$

**Calculation**

$H = -[(0.66 \times \log_2(0.66)) + (0.33 \times \log_2(0.33))]$
$\log_2(0.66) \approx -0.599$
$\log_2(0.33) \approx -1.599$
Therefore:
$H = -[(0.66 \times -0.599) + (0.33 \times -1.599)]$
$H = -[-0.3953 + -0.5277]$
$H \approx 0.923$

**Final Answer**

Entropy ≈ **0.92 bits**

**Interpretation**

The entropy value of approximately 0.92 bits indicates the amount of uncertainty in predicting the next word after "science". The probability of "is" is much higher than the probability of "drives", so the model has relatively high confidence in predicting "is". If the probabilities of possible next words were more evenly distributed, the entropy would be higher, indicating greater uncertainty. If one word had a probability close to 1, the entropy would be close to zero, indicating very high prediction confidence.

**Importance in Mobile Keyboard Prediction**

- **Low entropy** means the next word is highly predictable.
- **High entropy** means several next words are possible.
- Entropy helps measure the uncertainty of language-model predictions.
- A keyboard can use low-entropy predictions to provide more confident suggestions.
- Higher entropy indicates that the keyboard may need more context before making a reliable prediction.
- Therefore, entropy is useful for evaluating the quality and confidence of intelligent text-prediction systems.

## CASE STUDY 2: AI-Powered Customer Support Chatbot

A multinational airline deploys a customer support chatbot that processes user messages and automatically identifies the grammatical role of words before generating responses.

Consider the following user inputs:

1. "Book a flight ticket now."
2. "This book is interesting."

The chatbot uses the Penn Treebank POS Tagset and a Hidden Markov Model (HMM) for probabilistic tagging.

Given:

- P(book | VB) = 0.6
- P(book | NN) = 0.4
- P(Start → VB) = 0.5
- P(Start → NN) = 0.5

The system must determine whether the word "book" functions as a Verb (VB) or Noun (NN) depending on context.

### Questions

1. Assign appropriate Penn Treebank POS tags to every word in both sentences. Justify the tag assigned to the word "book" using contextual clues.
2. Using the HMM probabilities, compute the likelihood of "book" being tagged as a Verb (VB) in the sentence "Book a flight ticket now." Explain why the probabilistic model favors this tag.
3. Compare Rule-Based Tagging and Stochastic (HMM) Tagging in resolving lexical ambiguity. Recommend which approach is more suitable for large-scale chatbot deployment and justify your answer.
4. Evaluate the role of standardized POS tagsets and word classes in improving intent detection, response generation, and overall chatbot performance.

## 1. Penn Treebank POS Tagging

### Sentence 1

**"Book a flight ticket now."**

| Word | POS Tag | Meaning |
| --- | --- | --- |
| Book | VB | Verb |
| a | DT | Determiner |
| flight | NN | Noun |
| ticket | NN | Noun |
| now | RB | Adverb |

### Sentence 2

**"This book is interesting."**

| Word | POS Tag | Meaning |
| --- | --- | --- |
| This | DT | Determiner |
| book | NN | Noun |
| is | VBZ | Verb |
| interesting | JJ | Adjective |

### Contextual Justification

In **"Book a flight ticket now"**, "book" represents an **action** performed by the user. It means the user wants to reserve a flight. Therefore, **book = VB (Verb)**.
In **"This book is interesting"**, "book" refers to a physical or conceptual object. It is the subject of the sentence and is preceded by the determiner "This". Therefore, **book = NN (Noun)**.
Thus, the same word can have different POS tags depending on its surrounding context.

## 2. HMM Probability for "Book" as Verb

### Given

$P(\text{book} \mid VB) = 0.6$
$P(\text{book} \mid NN) = 0.4$
$P(\text{Start} \rightarrow VB) = 0.5$
$P(\text{Start} \rightarrow NN) = 0.5$

### Formula

The HMM likelihood for a tag can be calculated as:
$P(\text{book}, VB) = P(\text{Start} \rightarrow VB) \times P(\text{book} \mid VB)$

**Calculation**

$P(\text{book}, VB) = 0.5 \times 0.6$
$P(\text{book}, VB) = 0.30$
For comparison:
$P(\text{book}, NN) = P(\text{Start} \rightarrow NN) \times P(\text{book} \mid NN)$
$P(\text{book}, NN) = 0.5 \times 0.4$
$P(\text{book}, NN) = 0.20$
**Comparison**
$P(\text{book}, VB) = 0.30$
$P(\text{book}, NN) = 0.20$
Therefore:
**VB > NN**

**Answer**

The HMM favors **VB (Verb)** because the likelihood of "book" being a verb is **0.30**, whereas its likelihood of being a noun is **0.20**.

**Interpretation**

The transition probabilities from the start state are equal for VB and NN. Therefore, the decision is mainly influenced by the emission probabilities. Since:
$P(\text{book} \mid VB) = 0.6 > P(\text{book} \mid NN) = 0.4$
the model assigns a higher probability to the verb interpretation. This agrees with the sentence **"Book a flight ticket now"**, where "book" represents an action.

**3. Rule-Based vs HMM Tagging**

**Rule-Based Tagging**

Rule-based tagging uses predefined grammatical rules and dictionaries to assign POS tags.
For example:

- A word after a determiner such as "a" is commonly a noun.
- A word followed by an object can sometimes function as a verb.
- Contextual grammatical patterns can be used to identify word classes.

For **"Book a flight ticket now"**, a rule can identify "book" as a verb because it appears at the beginning of an imperative sentence and is followed by the object "a flight ticket".

**Advantages**

- Easy to understand.
- Simple to implement.
- Does not require a large training corpus.
- Rules can be manually controlled.

**Limitations**

- Requires many manually designed rules.
- Difficult to handle complex language.
- Rules may fail when sentences have unusual structures.
- Maintenance becomes difficult for large applications.

**Stochastic HMM Tagging**

An HMM uses probabilities learned from training data.
It considers:

- Emission probabilities.
- Transition probabilities.
- Contextual tag sequences.
- Likelihood of a word belonging to a particular tag.

For example:
$P(\text{book} \mid VB) = 0.6$
$P(\text{book} \mid NN) = 0.4$
Therefore, the model favors VB when the contextual probabilities support the verb interpretation.

**Recommendation**

For a **large-scale airline chatbot, stochastic HMM tagging is more suitable than a purely rule-based system** because it can learn grammatical patterns from large amounts of training data and handle lexical ambiguity more effectively.
However, a **hybrid approach** combining statistical tagging with selected linguistic rules would generally be stronger in practice. Statistical methods provide adaptability, while rules can handle important domain-specific patterns and exceptions.

**4. Role of POS Tagsets and Word Classes**

**Standardized POS Tags**

The Penn Treebank tagset provides standardized grammatical categories such as:

- **NN** = Noun
- **VB** = Verb
- **JJ** = Adjective
- **RB** = Adverb
- **DT** = Determiner
- **PRP** = Personal Pronoun
- **VBZ** = Third-person Singular Verb

Using standardized tags allows different NLP components to interpret grammatical information consistently.

### Impact on Intent Detection

POS information helps the chatbot identify the user's intended action.

For example:
**"Book a flight ticket."**
The word **"Book" = VB** indicates an action or request. This helps the chatbot identify a possible **flight-booking intent**.

In contrast:
**"This book is interesting."**
The word **"book" = NN** indicates an object rather than an action. Therefore, the chatbot should not interpret this as a flight-booking request.

### Impact on Response Generation

POS tags help the system understand sentence structure before generating a response.
For example:

- **VB** can indicate an action or request.
- **NN** can represent an entity or object.
- **JJ** can describe a property.
- **RB** can provide information about manner or time.

This grammatical information helps the response-generation component produce more relevant responses.

### Overall Benefits

- Improves grammatical understanding.
- Helps resolve ambiguous words.
- Supports accurate intent classification.
- Improves entity and action identification.
- Helps generate contextually appropriate responses.
- Provides a consistent representation for different NLP modules.

### Final Conclusion

POS tagging is important for the airline chatbot because the same word can have different grammatical meanings depending on context. In this case, **"book" is VB in "Book a flight ticket now" and NN in "This book is interesting."** Standardized Penn Treebank tags combined with probabilistic HMM tagging provide a structured way to resolve such ambiguity and improve intent detection and response generation.

## CASE STUDY 3: News Analytics and POS Tag Correction System

A digital news analytics company processes millions of news articles daily. The platform uses a Transformation-Based Tagger (Brill Tagger) to improve tagging accuracy after an initial POS tagging phase.

The sentence processed by the system is:

"Economic growth increases employment."

Initial POS Tags:

- economic/JJ
- growth/NN
- increases/NNS
- employment/NN

A learned transformation rule states:

**Change NNS to VBZ if the preceding word is tagged as NN.**

The system also performs corpus-based frequency analysis and uncertainty evaluation.

### Questions

1. Apply the transformation rule and update the POS tag sequence. Explain why the correction is linguistically appropriate.
2. Analyze the initial tagging errors and justify the corrected tag assignments using grammatical structure and sentence semantics.
3. Assuming the corpus contains the words:
    - economic (120 occurrences)
    - growth (450 occurrences)
    - increases (210 occurrences)
    - employment (380 occurrences)

    Compute the word frequency distribution and explain how frequency information supports probabilistic POS tagging.

4. Evaluate how Transformation-Based Tagging improves tagging accuracy compared with the initial tagging stage. Discuss how entropy can be used to measure uncertainty in tag assignments before and after rule application.

## 1. Transformation Rule Application

### Given Sentence

"Economic growth increases employment."

### Initial POS Tags

economic/JJ
growth/NN
increases/NNS
employment/NN

### Given Rule

**Change NNS to VBZ if the preceding word is tagged as NN.**

### Applying the Rule

The word **"increases"** is initially tagged as **NNS**.
The preceding word is **"growth"**, which is tagged as **NN**.
Therefore, the transformation rule applies:
NNS → VBZ

### Updated POS Sequence

economic/JJ
growth/NN
increases/VBZ
employment/NN

### Answer

The corrected sequence is:
**economic/JJ growth/NN increases/VBZ employment/NN**

### Interpretation

The correction is linguistically appropriate because **"growth"** is the subject of the sentence, while **"increases"** is the main verb. The sentence follows the structure:

**Adjective + Noun + Verb + Noun**

Here, "economic" describes "growth", "growth" acts as the subject, "increases" expresses the action, and "employment" is the object. Therefore, **increases/VBZ** is grammatically more appropriate than **increases/NNS**.

## 2. Analysis of Initial Tagging Errors

### Initial Tags

economic/JJ
growth/NN
increases/NNS
employment/NN

### Error Identified

The main error is:
**increases/NNS → increases/VBZ**
The tag **NNS** represents a plural noun, while **VBZ** represents a third-person singular present-tense verb.

### Grammatical Analysis

"economic" is correctly tagged as **JJ** because it describes the noun "growth".
"growth" is correctly tagged as **NN** because it functions as the subject of the sentence.
"increases" is incorrectly tagged as **NNS** because it does not represent a plural noun. It expresses an action performed by the subject "growth".
"employment" is correctly tagged as **NN** because it represents the object affected by the action.

### Correct Structure

**economic/JJ → growth/NN → increases/VBZ → employment/NN**

### Semantic Interpretation

The sentence means that **economic growth causes employment to increase**. The word "increases" therefore represents an action or state change, making **VBZ** the appropriate grammatical category.
The transformation rule successfully detects this contextual relationship and corrects the initial tagging error.

## 3. Word Frequency Distribution

### Given Frequencies

economic = 120
growth = 450
increases = 210
employment = 380

### Total Frequency

Total = 120 + 450 + 210 + 380
Total = **1160**

**Frequency Distribution**

| Word | Frequency | Probability |
|---|---|---|
| economic | 120 | 0.1034 |
| growth | 450 | 0.3879 |
| increases | 210 | 0.1810 |
| employment | 380 | 0.3276 |
| **Total** | **1160** | **1.0000** |

**Calculations**

P(economic) = 120 / 1160 = **0.1034**
P(growth) = 450 / 1160 = **0.3879**
P(increases) = 210 / 1160 = **0.1810**
P(employment) = 380 / 1160 = **0.3276**

**Interpretation**

The word **"growth"** has the highest frequency with approximately **38.79%**, followed by **"employment"** with **32.76%**. "Increases" has approximately **18.10%**, while "economic" has the lowest frequency at **10.34%**.
Frequency information is useful in probabilistic POS tagging because frequently occurring words provide stronger statistical evidence for estimating word-tag probabilities. For example, if "increases" frequently occurs with the **VBZ** tag in the training corpus, the tagger can assign a higher probability to VBZ when encountering the word in a similar context.
However, word frequency alone cannot always determine the correct POS tag because the same word can have different grammatical roles in different contexts. Therefore, frequency information should be combined with contextual and transition information.

**4. Transformation-Based Tagging and Entropy**

**Initial Tagging**

The initial tagger produces:
economic/JJ
growth/NN
increases/NNS
employment/NN

There is one clear tagging error:
**increases/NNS**

The transformation rule changes it to:

**increases/VBZ**

**Effect of Transformation**

Before transformation:
**economic/JJ growth/NN increases/NNS employment/NN**

After transformation:
**economic/JJ growth/NN increases/VBZ employment/NN**

The transformation-based tagger improves accuracy by examining the context surrounding incorrectly tagged words and applying learned correction rules.

**Advantages**

- Corrects errors made by the initial tagger.
- Uses contextual information.
- Learns transformations from training data.
- Improves grammatical consistency.
- Reduces common POS tagging errors.
- Is especially useful when words have multiple possible POS tags.

**Entropy and Uncertainty**

Entropy measures the uncertainty associated with possible POS tags.
The general formula is:
$H(X) = -\Sigma P(tag) \log_2 P(tag)$
If a word has two possible tags with similar probabilities, uncertainty is high and entropy is high.
For example:
$P(NNS) = 0.5$
$P(VBZ) = 0.5$
Then:
$H = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)]$
H = **1 bit**
This represents high uncertainty because both tags are equally likely.
After applying contextual information, suppose:
$P(VBZ) = 0.9$
$P(NNS) = 0.1$
Then:
$H = -[0.9 \log_2(0.9) + 0.1 \log_2(0.1)]$
H ≈ **0.469 bits**
The entropy decreases from **1 bit to approximately 0.469 bits**.

**Interpretation**

A decrease in entropy indicates that the tagger has become more confident about the correct POS tag.

Before transformation, "increases" could potentially be interpreted as either a noun or a verb. After considering the preceding **NN "growth"** and the sentence structure, the system strongly favors **VBZ**. Therefore, Transformation-Based Tagging improves both **tagging accuracy and confidence**, while entropy provides a quantitative measure of the uncertainty before and after correction.

## Rubrics for Calculation

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Understanding of Case | 25 | Thorough understanding; clearly identifies all key issues | Good understanding with most issues identified | Basic understanding; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25 | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps | Limited application of concepts | Fails to apply relevant concepts |
| Analysis | 20 | In-depth analysis with strong reasoning and insights | Good analysis with logical reasoning | Basic analysis with limited depth | Weak or no analysis; lacks reasoning |
| Solution Design | 20 | Innovative, feasible solutions with strong justification | Logical solutions with adequate justification | Basic solutions with weak justification | Incorrect or no solution provided |
| Clarity | 10 | Clear, well-structured, and easy to understand | Organized and understandable presentation | Limited clarity and structure | Poor clarity; difficult to understand |

| Q. No. / Criteria Level | Understanding of Case | Application of Concepts | Analysis | Solution Design | Clarity | Sub. Total | Total % |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |

| Faculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| | | |
| Signature: ________________ | Signature: ________________ | Signature: ________________ |
| Date: ____________________ | Date: ____________________ | Date: ____________________ |